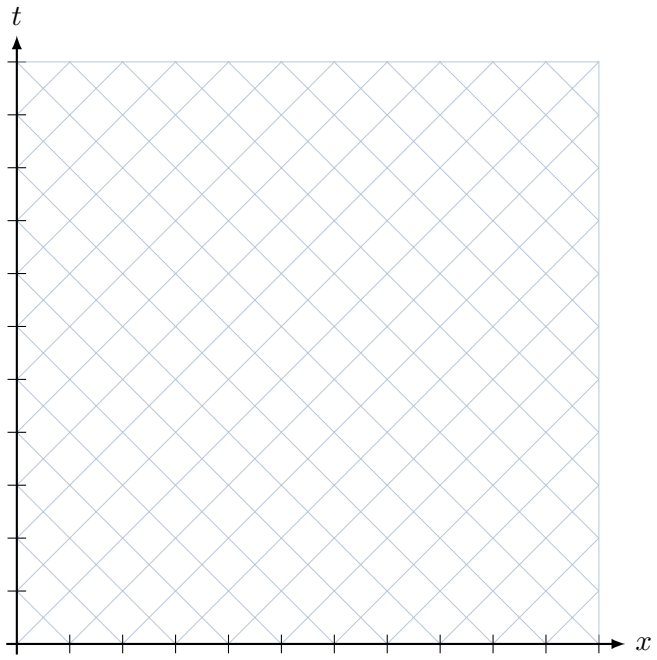
1. Mt. Hood and Mt. Rainier are 300 km apart in their rest frame. They erupt at the same time in the frame of a seismologist halfway between them. A spacecraft flying at $\beta = \frac{4}{5}$ in the direction from Mt. Rainier to Mt. Hood is directly over Rainier when it erupts.



A. Calibrate and label the worldlines and t = 0 nowslice of the Seismologist, Mt. Rainier, and Mt. Hood. Choose consistent units (e.g. seconds and ls, or miliseconds and mls). Label a few other nowslices.

B. Calibrate and label the worldline and t = 0 nowslice of the spaceship.

C. Label the following events R (Rainier Erupts), H(Hood Erupts).

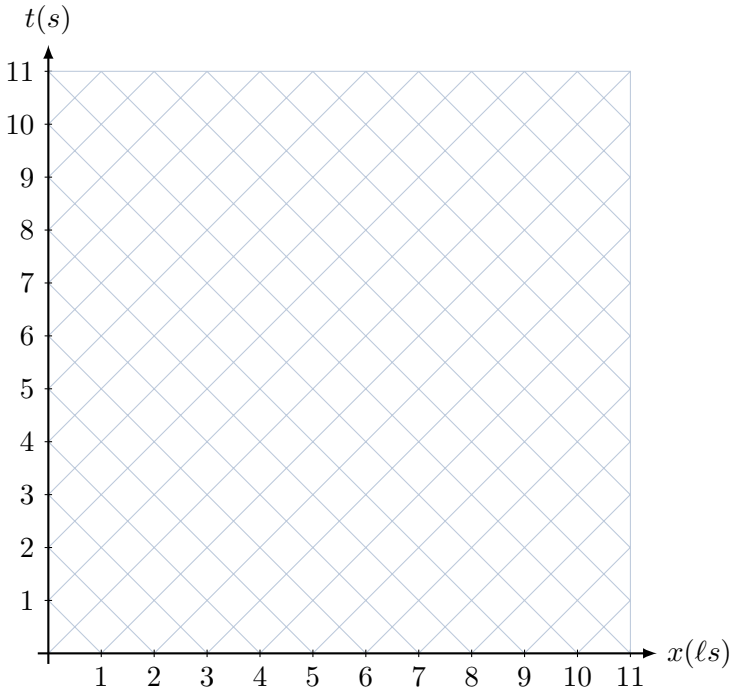
D. What are the coordinates (t, x) of Event R in the Mountain frame?

E. What are the coordinates (t, x) of Event R in the Ship frame?

F. What are the coordinates (t, x) of Event H in the Mountain frame?

G. What are the coordinates (t, x) of Event H in the Ship frame?

2. Jodi is running to the right at $\beta = \frac{3}{5}$ passes George. Their origins coincide at t = 0, this is event O.



A. Draw and calibrate Jodi's Worldline and t= 0 nowslice

B. Event F at (t,x) =(6.5,5.5) is when fireworks erupt in George's frame. Whenwhere does the firework erupt in Jodi's frame?

C. What is Δx and Δt between O and F in both frames

D. Show that Δs^2 is the same in both frames.

3. The Inexorable The federation ship Inexorable is floating in Federation territory at rest relative to the border of Klingon Space, which is 6 min away in the +x direction. Suddenly a Klingon warship flies past the Inexorable in the direction of the border at a speed of 3/5. Call this event A, and let this define the zero of time in both frames.

A. After 5 minutes, according to the Inexorable's clocks, the Klingon's emit a parting disrupter blast. Call this event B.

B. The disrupter beam travels at the speed of light back to the Inexorable' and disables it. This is event C.

C. Some time later the Klingons cross the boundary. This is event D. How much later is this in the Inexorable's frame? Now, the Klingon-Federation treaty states that it is illegal for a Klingon ship in Federation territory to damage Federation property. When the case comes up for adjudication, the Klingons claim that they are within the letter of the law, because in their reference frame, the damage occurred after they had crossed safely back into Klingon territory.

